SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE:CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving
Task

Code of Conduct:

I J. Kirthiga / 192421265 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: J. Kirthiga

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CO-4 ASSESSMENT TOOL-2

Network Design for 20-person Startup - Star topology

A Star topology is ideal for a small startup because all devices are connected to a central switch, ensuring easy management and troubleshooting. A 24-port layer 2/Layer 3 managed switch is used to connect all computers, printers, and VoIP devices. Each system is connected using Cat 6 UTP cables supporting Gigabit Speeds

The Switch is connected to router/firewall for Internet access, DHCP and NAT. A wireless access point extends connectivity for mobile devices

Justification:

- failure of one node does not affect others
- Easy fault isolation and maintenance
- Scalable for future Growth
- centralized management

Advantages:

- Easy troubleshooting and maintenance
- failure of nodes

Limitations:

- failure of switch affects entire network
- Slightly higher cost due to cabling

Core components Breakdown:

- central Switch, Cabling, Gateway, Wireless (WAP)

Wireless Access Point

2. Mesh Topology for Banking Networks

This is an excellent, highly accurate architectural Summary a high availability banking Network. It perfectly aligns with Enterprise financial Standards, balancing absolute resellience at the core with cost - effective redundancy at the edge.

Breakdown of critical architecture, protocols, configurations and hidden complexities required to make this work in production.

Core Routing and rapid convergence

While OSPF and BGP handle the rerouting, Standard timer settings are too slow for banking data

BFD - (Bidirectional Forwarding Detection) : Must be enabled on all mesh links. BFD detects link failures in milliseconds, triggering OSPF | BGP to reroute traffic before Standard protocol timers timeout

OSPF LSA Throttling - Tune OSPF timers for Sub-Second convergence ensuring link - State advertisements (LSAs) propagate instantly across the mesh.

Layer 2 - loop prevention & Gateway Redundancy

Because a mesh inherently creates physical loops, the layer 2 environment requires strict configuration.

Advantages:

- No single point of failure
- High reliability and redundancy
- Automatic rerouting of data
- Secure communication.

Justification:

Suitable for banking Systems where continuous operation is critical.

...oom Network Design - Bus Topology

...blem requirement - A classroom lab requires a low-cost ...work solution.

Major Technical corrections :

- Daisy - Chaining Via Hubs is Not a bus : - connecting computers through an unmanaged hub using Cat5e | Cat 6 Cable does not create a physical bus topology. It creates a physical star topology (or star-wired bus). Even though the hub broadcast data internally.

Switches create star topologies: Replacing terminators with an 8 - or - 16 port unmanaged switch does not achieve "bus like simplicity". A switch creates a true star topology. It completly isolates traffic to prevent data collisions which is the exact opposite of a bus Network.

Coaxial Cable is Obselete : True bus topology using RG-58 coaxial cables. BNC T- connectors, and 50- ohm terminators has been completly obselete for over two decades. finding Network interface cards (NIC's) that supports BNC connectors.

Advantages :

- Low cost
- Easy to Install
- Requires less cabling

Justification

Best Suited for Small labs with limited budget & low traffic.

~~Batl~~ Limitations :

- Backbone failure affects entire network
- Data collisions may occur
- Difficult fault detection

VLAN Provide...

4. University Campus Network - Hybrid Topology

A large campus requires Scalability, redundancy, and high performance.

Network components:

- Core routers and multilayer Switches
- Distribution Switches per building
- Access switches
- fiber-optic backbone (10-40 Gbps)
- Wireless access points (PoE)

Architecture Layers:

Core layer: High-Speed backbone

Distribution layer: connects buildings

Access layer: connects end devices

Key Advantages of this Design:

High redundancy: physical ring | mesh backbones ensure the campus stays online even if a main fiber link is cut

Seamless Scalability: New buildings connect directly to the core without disrupting existing Network traffic.

Cost Efficiency: combining Star Structures with localized bus/tree Setups balances high performance with lower Installation costs.

Advantages:

- High Scalable
- fault tolerant backbone
- Efficient Network Segmentation
- High Speed Communication

...design
...mplex design
...igh cost

...ication): Best for large Environments where Multiple topologies
...re required.

Multi-Department Company Network Design

...roblem requirement): Different Department need to Secure and
...icient communication.

...posed Design): Hierarchical network using VLAN-based segmentat-ion

...work components):
- ...yer 2 Switches (access Layer)
- ...yer 3 Switch/router (core layer)
- ...rewall
- ...ructured Cabling

...s layer): Each department, Such as HR, finance, Sales, and IT ...onnected to dedicated Layer 2 Switches. Those Switches connect ...users, printers and other devices within each department

... Configuration)

...ated VLANs are created for each department.

- ... - VLAN 10
- ...ance - VLAN 20
- ...es - VLAN-30
- IT - VLAN 40.

VLAN provides logical seperation, improve security and reduce unnecessary broadcast traffic.

Core layer: All department switches are connected to Layer 3 Switch or router, which acts as the core of the Network. It performs inter-VLAN routing allowing controlled communication between different departments

Internet Connectivity: A router connects the company's internal network to the internet or other external networks. A firewall can be used along with the router to protect the internal network from unauthorized access.

IP Addressing and Subnetting: Each department is assigned a seperate IP Subnet. proper IP addressing makes network management easier and helps control communication between departments

Scalability and Security: The hierarchical design makes the Network easy to manage, secure, troubleshoot and expand. New departments, users and devices can be added without significantly changing the existing Network.

Justification:

Therefore a hierarchical Network using layer 2 switches, VLANS, Layer 3 routing, firewall protection, redundancy and proper IP addressing is suitable for multi-department company because it provides better performance, security, reliability, manageability.